Swati Aggrawal

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PROFESSIONAL SUMMARY

MSc Computer Science graduate from Sapienza University of Rome specializing in Artificial Intelligence and Machine Learning. Experienced in machine learning research, NLP, Generative AI, and Computer Vision through academic research and applied projects. Skilled in developing and evaluating deep learning models using PyTorch and TensorFlow, designing experimental pipelines, and analyzing model performance. Published IEEE researcher with experience in transformer-based models, statistical analysis, and AI system development. Passionate about advancing AI research and building practical AI solutions.

SKILLS

- **Programming :** Python, Java, SQL, C++, Bash
- **Machine Learning & AI :** PyTorch, TensorFlow, Keras, Scikit-Learn, Deep Learning, Computer Vision, Generative AI, Vision Transformers, Continual Learning, Transfer Learning, Hugging Face Transformers, Supervised Learning, Unsupervised Learning, Hyperparameter Tuning, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Sentence Transformers, Transformer Models
- **Data Science & Analytics :** Pandas, NumPy, Matplotlib, Data Analysis, Statistical Analysis, Feature Engineering
- **Backend & APIs :** FastAPI, REST APIs
- **Model Risk & Analytics :** Model Development, Model Validation, Model Evaluation, Performance Benchmarking, Statistical Analysis, Predictive Modeling
- **Tools:** Git, GitHub, Linux, Docker, AWS

EDUCATION

Sapienza Università di Roma **May 2026**

Master of Science in Computer Science

- **Relevant Coursework:** Natural Language Processing, Deep Learning and Applied AI, Machine Learning, Computer Vision, Distributed Systems
- **Thesis:** Evaluating Pretrained Vision Transformers Under Continual Learning.
- Selected for a competitive international master's program.

Sharda University **July 2023**

Bachelor of Technology - CSE with AI and ML

- **Relevant Coursework:** Operating Systems, Data Structures, Machine Learning, Databases, Cloud Computing
- **Thesis:** Virtual Generative Model Inducing Human Identity to Computers in Metaverse
- Awarded **merit-based scholarship** (Top 5–10% of cohort).

PUBLICATION

Machine Learning for Air Pollution – IEEE Conference Publication (2023)

- Evaluated Random Forest, SVM, Decision Tree, and Neural Network models on environmental datasets.
- Performed comparative analysis using accuracy, robustness, and generalization metrics.
- Authored and presented findings through an IEEE peer-reviewed publication.

RESEARCH EXPERIENCE

Evaluation of Pretrained Vision Transformer Architectures Under Continual Learning Settings (Master's Thesis) **Nov 2025 - May 26**

- Conducted research on continual learning for Vision Transformer architectures, evaluating model robustness, catastrophic forgetting, and knowledge retention across sequential learning tasks.
- Reviewed and analyzed recent research literature on continual learning and transformer-based computer vision models to guide experimental design.
- Designed experimental pipelines for training, evaluation, and comparison of computer vision models across sequential tasks.
- Evaluated model performance under changing data distributions and analyzed model reliability.
- Investigated catastrophic forgetting and knowledge retention in transformer-based visual learning systems.

- Developed reproducible training and evaluation pipelines using Python and PyTorch.
- Automated experiment tracking and performance benchmarking across multiple continual learning scenarios.
- Documented experimental findings and identified research opportunities for future development of continual learning systems.
- **Technologies:** Python, PyTorch, Vision Transformers, Computer Vision

PROJECTS

DocDelta AI-Powered Document Intelligence & Comparison Platform

- Developed an AI-powered document intelligence application with backend APIs for automated document analysis and semantic comparison.
- Integrated Sentence Transformer models into inference pipelines for semantic similarity and document intelligence tasks.
- Designed RESTful endpoints using FastAPI for automated document analysis and change tracking workflows.
- Built document ingestion, semantic analysis, and comparison pipelines using transformer-based NLP models.
- Leveraged transformer-based embeddings for semantic document understanding and retrieval-oriented workflows.
- Implemented semantic matching to identify added, removed, and modified content across document versions.
- Containerized the application using Docker and deployed it on AWS ECS Fargate.
- Configured application monitoring and logging using AWS CloudWatch.
- **Technologies:** Python, FastAPI, Sentence Transformers, Docker, AWS ECS Fargate, CloudWatch, REST APIs

Metal Defect Detection using Computer Vision

- Developed computer vision models for automated defect detection using CNN, SVM, and Random Forest.
- Built preprocessing, training, and evaluation pipelines.
- Compared deep learning and traditional ML approaches.
- Performed model evaluation using accuracy, confusion matrices, and performance analysis.
- **Technologies:** Python, TensorFlow, Keras, OpenCV, Scikit-Learn

Lossy Text Compressor [Semantic NLP Pipeline]

- Investigated semantic text compression using transformer-based embeddings, quantization techniques, and text reconstruction methods.
- Evaluated reconstruction quality using cosine similarity and ROUGE metrics.
- Analyzed trade-offs between compression efficiency and semantic preservation across multiple quantization levels.
- **Technologies:** Python, PyTorch, Transformer Models, NLP, Sentence Embeddings